

16.6 Summary and Notes

16.6.1 Summary

Idea.

1. In a balanced layout every projection of interest is a Kronecker product of averaging matrices $\bar{\mathbf{J}}$, centring matrices $\mathbf{C}$ and identities. Products that differ by $\bar{\mathbf{J}}$ against $\mathbf{C}$ in some position are orthogonal, and ranks multiply (Lemma 16.1.1).
2. The additive two-way model has projection $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$. Row contrasts are estimated and judged as in a one-way layout that ignores the columns. With replication, the residual splits into interaction (lack of fit) and pure error. With one observation per cell, Tukey's one-degree-of-freedom test is an exact test against the kind of nonadditivity a wrong scale produces, and its slope suggests a power transformation (Theorem 16.1.2).
3. Interaction is a property of the table of cell means. It is present iff some interaction contrast (a zero-margin combination of cell means) is nonzero, and it has $(a-1)(b-1)$ dimensions whatever parameterization is used (Definition 16.2.1, Proposition 16.2.1).
4. An increasing transformation can remove interaction only if the orders of the levels are the same everywhere. Qualitative (crossing) interaction cannot be removed. Main effects in the presence of interaction are weighted averages of simple effects, and their values, even their signs, depend on the weights (Proposition 16.2.2, Proposition 16.2.3).
5. The balanced two-way layout with replication splits into grand mean, A , B , AB and error, with ranks 1, $a-1$, $b-1$, $(a-1)(b-1)$, $ab(m-1)$. The F tests are exact, and the sequential, Type II and Type III sums of squares all coincide (Theorem 16.3.1).
6. Products of orthogonal contrasts split the interaction sum of squares into single degrees of freedom. Tukey's method covers the levels of one factor, Scheffé's all interaction contrasts, and Bonferroni or Holm planned families (Proposition 16.3.2).
7. A balanced factorial with t factors has one term for each subset of factors, with projection $\mathbf{P}_S$ built from centring matrices in the positions of S . Hierarchical models have order-free sums of squares, and $\mathbf{P}_S \mathbf{y}$ is an alternating sum of marginal means (Theorem 16.4.1). The three-factor interaction measures how a two-factor interaction changes across the third factor (Proposition 16.4.2).
8. In a 2^t factorial every term has one degree of freedom, and its effect is a difference of two means with variance $4\sigma^2/n$ (Proposition 16.4.3). Pooling terms into error is exact if they are null and chosen in advance, and conservative if they are chosen in advance but not null. It is invalid if they are chosen because they look small (Proposition 16.4.4).
9. A nested factor has no main effect. Its sum of squares is the crossed B plus AB , the variation among inner levels within each outer level (Definition 16.5.1, Theorem 16.5.1). When the inner levels are a sample, the outer factor must be tested against the inner mean square, not against the within-level error.

16.6.2 Notes and sources

Sources. The coverage of this chapter was checked against Rencher and Schaalje (2008, chapter 14), Christensen (2020, chapter 7) and Seber and Lee (2003, sections 8.4–8.6). The projection approach follows the vector-space treatment of Christensen (2020), which develops interaction contrasts and their product construction (Proposition 16.3.2) with particular care. Rencher and

Schaalje compute the expected mean squares directly, and projections replace that computation here. Scheffé (1959) remains the classical account of fixed-effects multiway layouts, including main effects and interactions defined with general weights (compare Proposition 16.2.1(v)).

Factorial experiments. Fisher (1935) argued for factorial experiments over one-factor-at-a-time trials: every observation contributes to every main effect, and interactions can be studied at all. Yates (1937) developed the analysis of 2^t and 3^t designs, including the algorithm for 2^t effects that carries his name. Box, Hunter and Hunter (2005) and Wu and Hamada (2009) are standard modern texts.

Tukey's test. The one-degree-of-freedom test is due to Tukey (1949). Its exact distribution follows from the theorem of Milliken and Graybill (1970) on regressors computed from the fitted values (Theorem 14.7.2), because Tukey's regressor is, after the additive model is projected out, the table of squared fitted values; Exercise (C1) made the same connection. The power rule Eq. 16.1.5 is a heuristic, and Chapter 22 treats transformations systematically.

Interaction and scale. The dependence of interaction on the scale of measurement, and the order condition of Proposition 16.2.2, belong to the theory of conjoint measurement of Luce and Tukey (1964). The counterexample of Exercise (C1) is an instance of their double cancellation condition. In clinical trials a treatment effect that changes sign across patient subgroups is called a *qualitative interaction*, and Gail and Simon (1985) gave a likelihood ratio test for it.

Unreplicated factorials and pooling. The half-normal plot is due to Daniel (1959), and the pseudo standard error of Exercise (C1) to Lenth (1989). The effect of preliminary tests on subsequent inference was first studied by Bancroft (1944), and the simulation of Figure 16.4.2 illustrates his conclusions in a simple case.

Nested designs. Hurlbert (1984) named *pseudoreplication*, treating observations within an experimental unit as independent replicates of the unit's treatment, and documented how widespread it was in ecology. Random inner levels are taken up with variance components in Chapters 32 and 33. Unbalanced multiway data are the subject of Chapter 17, and randomized blocks and Latin squares, additive layouts whose structure comes from randomization, of Chapter 18.

16.6.3 References

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